

Yet, today, even if you check your mail on a smartphone, you get back to your laptop for jobs that can only be done on your desktop/notebook PC. The handheld today is striving hard to be a replacement, but is at best a complement.

Imagine, if you were able to work on your handheld as if it were your PC. If you were at home, you could simply log out of the handheld and log in to the home entertainment laptop. At office, you could work on a thin client and if, by chance, your handheld ran out of battery while travelling, you could continue from where you left off on a friend's handheld or from any other device capable of connecting to your data centre over the Internet.

If the above seems too futuristic, this article is for you. If it sounds stale, maybe the next article will be the one for you. Either way, you cannot escape the promise of desktop virtualisation; my desktop—everywhere, everytime.

But what is this new thing about desktop virtualisation? Have we not been using the remote desktop, VNC and other tools for ages? Have we not logged on to remote PCs and achieved the above? Ask all software developers and they will tell you how difficult it is to work without the ability to install/use their favourite editor/debugger. Ask any office assistant to print simple documents on remote desktops to local printers, or ask a sysadmin to copy a required patch from a local USB pen drive to the remote desktop, and they will tell you it's still not an easy task.

Desktop virtualisation is significantly different from server-based computing and that is why it holds great promise—especially for the BPO and software development (IT and ITes) verticals to begin with. Once the technology matures further, I am sure that it would be the platform of choice for all. Software as a Service (SaaS) might give way to the Desktop as a Service (DaaS).

To quote Wikipedia (en.wikipedia.org/wiki/Desktop_virtualization), "According to a report by Gartner, hosted services accounted for more than 500,000 desktop units as of March 2009, but will grow to 49 million desktop units by 2013, and may make up as much as 40 per cent of the worldwide professional personal computer market, by revenue."

Server-based computing (or SBC) refers to traditional methods of remote access—VNC, rdesktop, etc. Desktop virtualisation has distinct advantages over server-based computing for individual access. Some of these that can be clearly envisaged at this stage of the emerging technology are:

- A user experience similar to the local machine.
- Desktop virtualisation provides access to local audio, USB, etc, providing unparalleled ease of use and flexibility.
- CPU and GPU intensive tasks can be offloaded to the client instead of increasing the load on the central server. A user running a CPU-intensive compilation can slow down the central server for others.
- Complete customisation and freedom—all users can add/remove software, and customise the environment as per their preferences.

Apart from server-based computing, desktop virtualisation

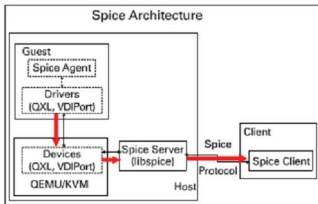


Figure 1: The SPICE architecture

has tremendous benefits over the 'desktop' model as well—one machine per user:

- Freedom from hardware crashes: In case the client hardware crashes, the remote session can be easily accessed from any other device.
- Better security: Again, this is based on the assumption that a data centre is generally more secure in all aspects.
- For a sysadmin, managing hundreds of individual machines versus managing virtual machines translates to ease of patch management and new deployments... the list is endless.

In short, desktop virtualisation is here to stay. It is still early days, but it is undoubtedly the way forward in these times of server virtualisation and cloud infrastructure.

There are already a few products in the market for desktop virtualisation, but it is the open source SPICE technology that holds the key, as far as I am concerned. Preventing vendor lock-ins and based on KVM (stock Linux kernel), it holds interesting prospects. Red Hat has open sourced the SPICE protocol that it acquired from Qumranet into Red Hat Enterprise Virtualisation for Desktops (currently in beta).

SPICE stands for Simple Protocol for Independent Computing Environments. The basic components of SPICE are:

1. The SPICE server
2. The SPICE client
3. The SPICE protocol

In a traditional SBC environment, there are just two layers—the back-end and the front-end. The back-end layer resides on the server host, and the front-end layer on the client. The front-end layer of the software interprets the inputs—the keys' input and mouse movements—and transmits them over to the back-end. The backend layer exposes the graphical display and the front-end layer, typically, plugs into this.

How SPICE operates differently is that the backend layer does not connect directly to the underlying hardware through the operating system. The back-end layer is a hardware emulation layer. The desktop to be logged in is actually a virtualised guest that exposes itself through the Virtual Device Interfaces (VDI), such as a virtual graphics driver, etc.

In the middle of the back-end and the remotely connecting client sits the SPICE server, which now actually becomes the front-end for the virtual devices.